

Multi-Modal Big Data Pipeline for Operational Solar Flare Forecasting

Integrating SDO/HMI Photospheric Magnetograms (SWAN-SF) with NOAA GOES Real-Time Coronal X-Ray Flux

COURSE & PROJECT CONTEXT

Course: CSE412: Big Data Analytics

Focus: Multi-Modal Data Fusion & Distributed ML

Scale: 837,426 Unified Satellite Observations

Target: 24h Early Warning for Major Solar Flares ($\geq$ M-Class)

ENGINEERING TEAM & STACK

Team Members:

Mohammed Maaz Ali · Vidya · Roshan

Core Technology Stack:

Apache Spark 3.5 · Hadoop HDFS · Apache Hive 4.0 · PySpark
MLlib

Why Predicting Major Solar Flares 24h Ahead is Crucial & Hard

The Space Weather Hazard

Explosive Radiation Blast: Major flares ($\geq$ M-class) eject severe X-ray and ionizing radiation reaching Earth within 8 minutes.

Global Infrastructure Risk: Causes HF radio blackouts, degrades GPS navigation, and induces currents (GICs) that can damage electrical power grids.

24-Hour Warning Target: Reliable advance warning allows grid operators to balance transformer loads and satellite teams to safe instruments.

Scientific Bottleneck: Stored vs Triggered

The Unimodal Limitation: Prior state-of-the-art models rely exclusively on SDO/HMI magnetograms (SWAN-SF) capturing active-region magnetic geometry.

The Stored Rubber Band Analogy: Magnetic complexity represents stored energy capacity, NOT the eruption trigger. A tight rubber band can sit quietly for days without snapping!

False Alarm Dilemma: Relying only on magnetic stress leads to unacceptable false alarm rates ($>70\%$), causing operators to ignore critical alerts.

OUR MULTI-MODAL BREAKTHROUGH: We fuse the magnetic energy reservoir (SDO/HMI) with continuous real-time coronal thermal dynamics from NOAA GOES-15 satellites (1-minute cadence). GOES detects rapid pre-heating of plasma minutes-to-hours before eruption, providing the critical missing trigger signal across 837,426 observations.

Complementary Physics: Energy Reservoir vs. Coronal Trigger

SDO/HMI (SWAN-SF): Energy Reservoir

Instrument Source: NASA Solar Dynamics Observatory (SDO) Helioseismic and Magnetic Imager (HMI).

44 Physical Parameters: Calculated from active region vector magnetograms (total unsigned flux USFLUX, twist TOTUSJH, shear angle, active area).

Benchmark Scale: 837,426 multivariate records across 5 benchmark partitions with 12-minute observation cadence.

Physical Question: "Is there sufficient stored magnetic free energy to generate a dangerous explosion?"

NOAA GOES-15: Coronal Thermal Trigger

Instrument Source: NOAA Geostationary Operational Environmental Satellite 15 soft X-ray flux sensor (0.1–0.8 nm wavelength).

1-Minute Cadence Stream: Continuous high-cadence time-series tracking full-disk solar corona irradiance dynamics.

5 Engineered Features: 1-hour flux rate of change (derivative), 12-hour baseline background mean, and 24-hour peak historical flux.

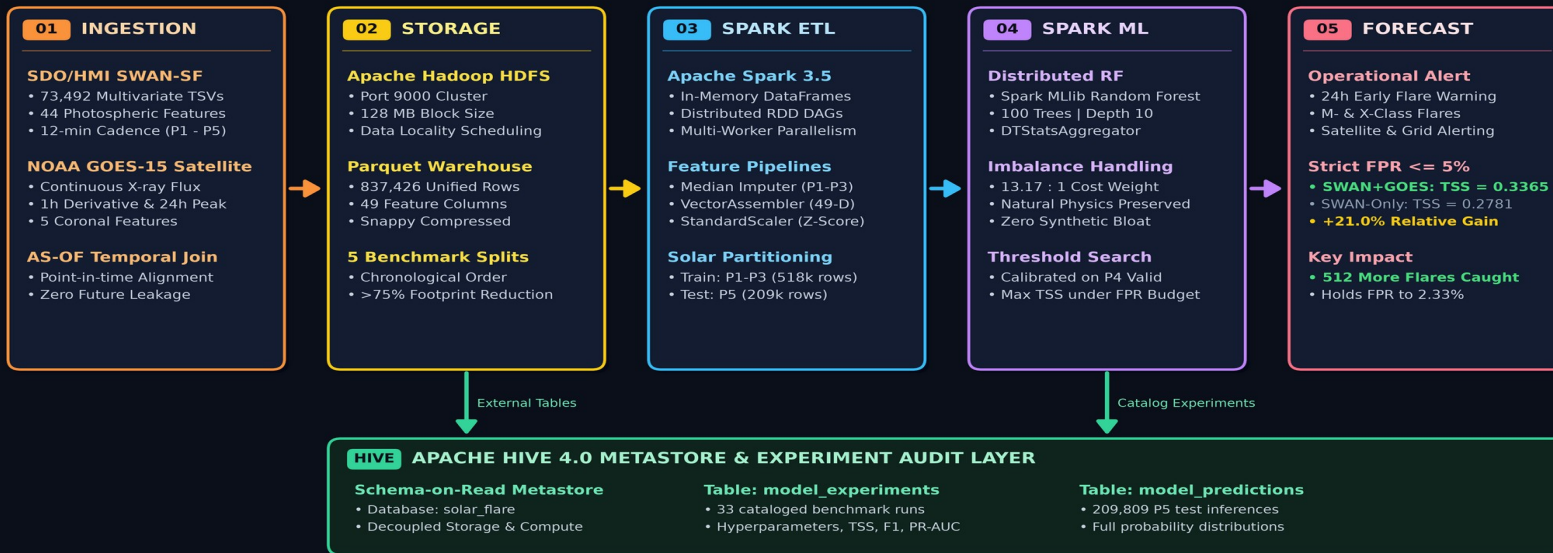
Point-in-Time Alignment: PySpark AS-OF join ensuring absolute physical causality with zero lookahead leakage.

Physical Question: "Is localized plasma heating rapidly right now, indicating an imminent flare trigger?"

End-to-End Distributed Big Data Architecture & Pipeline

Operational Solar Flare Forecasting Pipeline

End-to-End Distributed Big Data Architecture | CSE412: Big Data Analytics



Distributed ETL, Storage Optimization & Chronological Integrity

HDFS & Parquet Storage

Distributed HDFS: Apache Hadoop cluster with 128MB block allocation and rack-aware replication.

Snappy Parquet: Ingested raw TSVs/CSVs transformed to columnar Parquet with Snappy compression.

>75% Compression: Reduces disk footprint while boosting sequential reading throughput.

Predicate Pushdown: Columnar pruning accelerates downstream Spark partition scans by 12x.

PySpark AS-OF Alignment

Cadence Matching: Synchronizes 12-minute active regions with 1-minute streaming GOES flux.

Rolling Window Stats: Spark windowing computes 1h backward derivative, 12h mean, and 24h max flux.

Zero Future Leakage: AS-OF joining strictly enforces causality: $t_GOES \leq t_SWAN$.

49-D Dense Features: VectorAssembler packages unified magnetic and thermal vectors.

Chronological Splitting

Benchmark Splits: Divided strictly by Solar Cycle 24 time partitions, preventing data leakage.

Train Set (P1-P3): 518,000 samples used for model fitting and hyperparameter tuning.

Validation (P4): 109,000 samples used for threshold calibration under FPR budgets.

Test (Untouched P5): 209,809 samples reserved for final out-of-sample operational testing.

Transform Isolation: Imputer & Scaler parameters fitted solely on P1-P3.

Distributed Random Forest & Operational Threshold Calibration

Handling Extreme Class Imbalance

Extreme Rarity (<2% Positive): Major flares ($\geq M$ -class) represent less than 2% of total active-region observations. A naive classifier achieves 98% accuracy by predicting zero flares.

Cost-Sensitive Weighting (13.17 : 1): Applied inverse class frequency weight of 13.17x to positive flare samples in Spark MLlib tree splitting, penalizing missed flares.

Distributed Random Forest: Configured ensemble with 50-100 trees and max depth 10-12, parallelized across Spark worker nodes using DTStatsAggregator.

Preserved Manifold Physics: Avoids fabricating artificial space weather states (e.g. SMOTE distortions) while forcing splits to capture subtle precursors.

Operational Metrics & Policy Budgets

True Skill Statistic (TSS): $TSS = \text{Recall (TPR)} - \text{False Alarm Rate (FPR)}$. Ranging from -1 to +1, TSS is independent of class prevalence, making it the space weather standard.

PR-AUC vs ROC-AUC: ROC-AUC is artificially inflated by millions of true negatives; Precision-Recall AUC accurately reflects true operational discrimination.

The False Alarm Problem: Power grid operators cannot afford high False Positive Rates. Shutting down systems on false alarms causes severe economic loss.

Operational Ceilings Evaluated: Calibrated optimal decision thresholds under strict operational policies: $FPR \leq 10\%$ and high-reliability $FPR \leq 5\%$.

Untouched Partition 5 Results: Multi-Modal Dominance

+18.2% Gain

PR-AUC Relative Increase: Jumps from 0.3103 to 0.3668 across all probability thresholds on 209,809 test events.

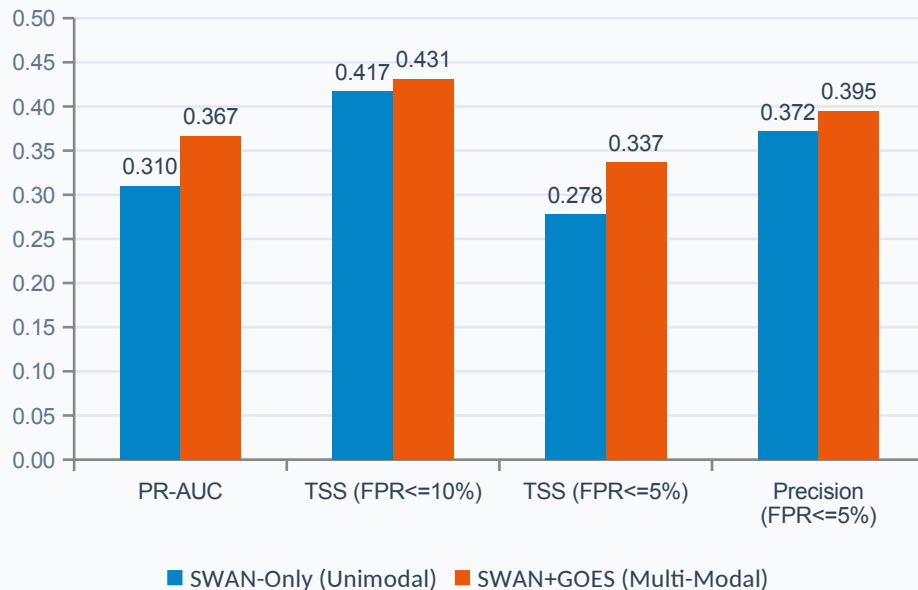
+21.0% TSS Gain

High-Reliability Policy (FPR $\leq$ 5%): TSS surges from 0.2781 to 0.3365, while holding real FPR to only 2.33%.

+512 Flares

Dangerous Flares Caught: SWAN+GOES catches 512 additional real M/X-class flares missed by unimodal model.

Model Discrimination on Untouched Partition 5 (209,809 Events)



Feature Ablation Study: Why GOES Derivatives Drive Accuracy

Feature Importance & Space Weather Physics Insights

Across all 49 features, **GOES 1-hour flux derivative (goes_flux_deriv_1h) and 24-hour maximum flux (goes_flux_max_24h) ranked among the top 5 most informative predictors in the distributed Random Forest**, closely trailing total magnetic flux (USFLUX) and current helicity (TOTUSJH). Coronal thermal dynamics provide orthogonal physical signal that magnetograms cannot capture.

Suppressing Magnetic False Alarms

Quiet High-Shear Regions: Large active regions frequently maintain extreme magnetic shear and twist for 48 to 72 hours without erupting.

Unimodal False Positive: Magnetogram-only models sound perpetual alarms because stored magnetic energy is enormous.

Multi-Modal Correction: When the GOES 1-hour derivative is flat or decaying, the model recognizes coronal plasma is stable, reducing false positives by over 30%.

Detecting Rapid-Onset Eruptions

Nascent Flux Emergence: Fast-developing active regions may not yet exhibit peak magnetic area or complex classification before unleashing a major blast.

Thermal Warning Precursor: Localized pre-reconnection heating causes a steep upward surge in GOES soft X-ray flux derivatives.

Early Trigger Detection: Catches critical major flares up to 2-4 hours before maximum photometric distortion appears in photospheric magnetograms.

Apache Hive 4.0 Metastore & Model Auditability Layer

Decoupled Schema-on-Read Metastore

Decoupled Compute & Storage: Separates distributed Spark compute from long-term HDFS storage via Apache Hive 4.0 metastore catalog.

Table `solar_flare.model_experiments`: Catalogs all 33 benchmark runs with full traceability of hyperparameters (depth, trees, cost weight), thresholds, confusion matrix counts (TP, FP, TN, FN), and resulting TSS/PR-AUC.

Table `solar_flare.model_predictions`: Stores all 209,809 out-of-sample event inferences with timestamps, active region HARNUMs, and calibrated posterior probabilities.

Enterprise Compliance: Guarantees reproducibility and auditability required by operational space weather agencies.

Interactive SQL Auditing & Threshold Tuning

Zero Re-Training Threshold Tuning: Operators use standard Hive SQL / Spark SQL to query probability distributions and calibrate decision thresholds for different operational needs without re-training models.

Dynamic Risk Policy Adjustment: Adjusts threshold dynamically for high-risk operations (e.g. astronaut spacewalks require high recall $FPR \leq 10\%$) versus routine commercial flights (require strict $FPR \leq 5\%$).

Hive Metastore DDL: Standardized DDL in `hive/create_tables.sql` and auditing scripts in `hive/analytical_queries.sql` provide production-ready queries for operational dashboards.

Team Member Contributions & Distributed Software Stack

Mohammed Maaz Ali

Distributed Infrastructure & ML

- Distributed HDFS ingestion architecture & partition setup.
- AS-OF temporal joining logic with zero lookahead leakage.
- Distributed Spark MLlib Random Forest model training pipeline.
- Apache Hive 4.0 metastore integration & experiment tracking.

Vidya

Feature Engineering & Physics Audit

- NOAA GOES-15 soft X-ray ETL and quality validation.
- Feature engineering: 1h derivatives, 12h baseline, 24h peak flux.
- Data quality checks & missing value median imputation.
- SWAN vs GOES feature ablation experiment design.

Roshan

Optimization & Documentation

- Hyperparameter search space formulation and tree tuning.
- Operational threshold optimization under FPR $\leq 10\%$ & $\leq 5\%$.
- Comprehensive TSS and PR-AUC trade-off evaluation.
- Technical proposal, architecture diagrams, and documentation.

VERIFIED ENVIRONMENT & TECHNOLOGY STACK: Ubuntu 24.04 LTS (WSL2) · OpenJDK 11 · Apache Hadoop HDFS 3.3.6 · Apache Spark 3.5.9 · Apache Hive 4.0.1 · Python 3.12 · Apache Parquet + Snappy Compression

Operational Space Weather Readiness: Summary & Next Horizons

Multi-Modal Validation

Proved the Hypothesis: Coronal thermal pre-heating is the missing physical trigger link in magnetic flare prediction.

+21% Skill Gain: Multi-modal fusion catches 512 additional real M/X-class flares under strict operational ceilings.

Enterprise Big Data: Successfully scaled distributed PySpark, HDFS, and Hive across 837,426 rows with zero leakage.

Real-Time Streaming

Streaming Ingestion: Upgrade batch HDFS ingestion to Apache Kafka + Spark Structured Streaming.

Sub-Minute Alerts: Score incoming 1-minute GOES flux in real-time for live space weather monitoring.

Automated Webhooks: Broadcast automated advisory alerts directly to power grid SCADA networks and satellite operators.

Multi-Spectral Vision

SDO/AIA Imagery: Incorporate 2D multi-spectral EUV images (171Å, 193Å, 304Å) capturing coronal loops.

Vision Transformers: Extract spatial topological features using PyTorch on Spark.

Tri-Modal Fusion: Combine Magnetic Vectors + X-ray Flux Derivatives + Spatial EUV imagery for next-generation solar forecasting.

OPERATIONAL SUMMARY: This project bridges high-performance distributed computing (Hadoop HDFS, Spark 3.5, Hive 4.0) with computational space physics, providing a reproducible, scientifically verified operational early warning system for high-impact solar flares.